

29th Jan 2021 | CS6370

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⇒ IR Basics

- query → tokenization, spell check, lemmatization, stopword removal
- cons of bottom up → large amount of data, sensitive to the representation of the data.

Documents

↓
Pre-processing
↓
Inverted index

Query

↓
Pre-processing

- Computational linguistics ≠ NLP (see even domain experts)
- there is no strict rule as to when tokenization is to be done → can be done, before or after post-processing as it suits the task.
- Turing Omnibus → impt. book that has
 ↳ collection of articles
 ↳ understand the time complexity
- put ranking to the documents based on the query.
- retrieve from the inverted index (Pastings file)
 word → documents where it occurs one by one
 { }
 ↳ then it is easier to find the union

$$D \cdot A = (D, A) \text{ matrix}$$

$$|A| \cdot |A|$$

(inner product)

—/—/—

- Time complexity analysis

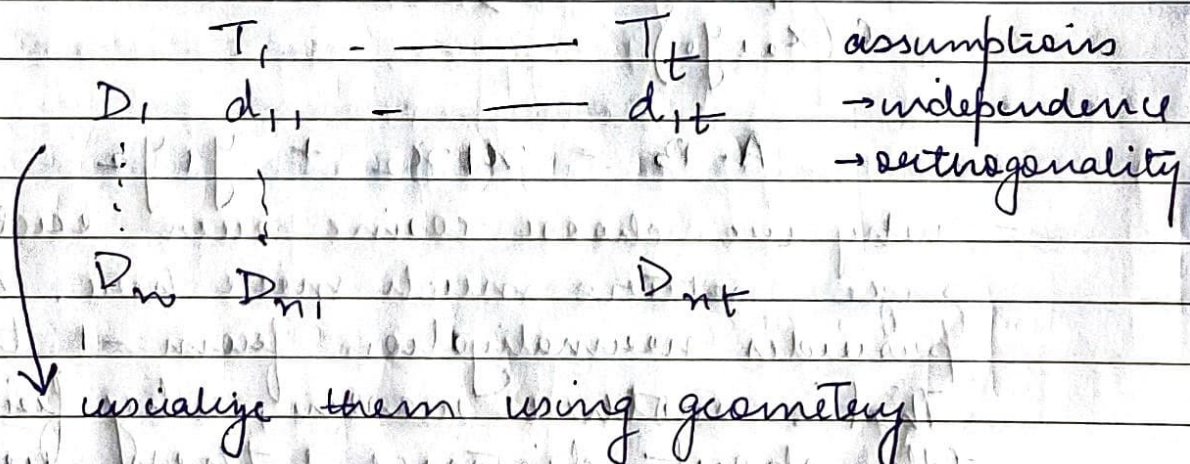
$$O(n^2)$$

- Now, we come to evaluate measures that tells how good our search engine.

- Process

a. markup free doc. text

c. weighting the words



visualize them using geometry

D_1 & D_2 is more similar to Q ?

What are you checking →

similarity? distance?

cosine similarity ✓ angle? projection?

$$\text{sim}(A, B) = \frac{A \cdot B}{|A| \cdot |B|} = \cos \theta$$

(local measure)

- term frequency

$$D = 2T_1 + 3T_2 + 5T_3$$

$$A = 0T_1 + 0T_2 + 2T_3$$

$$D_2 = 3T_1 + 1T_2 + 1T_3$$

IDF was global measure.

$$tf \times idf = w = tf \times \log\left(\frac{N}{n}\right)$$

- concept of dot prod. $\rightarrow$ term weights

$$\begin{matrix} \uparrow \\ \cdot A(x_1, y_1) \\ \cdot B(x_2, y_2) \\ \cdot G(x_0, y_0) \end{matrix}$$

$$A \cdot B = x_1 x_2 + y_1 y_2$$

= why we choose cosine over euclidean? angle matters much more here. Also, it provides normalization from -1 to +1.

That's another reason, why first we talk about distance & then we talk about similarity. There is a particular reason, why it is imp. to deal with "direction" and not magnitude.

Query is quite small as compared to documents. $\therefore$, euclidean distance does not make much sense.

$$\sin(A, B) = \cos \theta = \frac{A \cdot B}{|A| \cdot |B|}$$

$$W = tf \times idf = tf \times \log\left(\frac{N}{n}\right)$$

→ term weighing mechanism

- eq. of table given

✓ build your own IR system.

✓ TF x IDF (why it is multiplication)

← feature selection vs extraction

how
they are
related
to each other

→ Feature selection

- it is the process of choosing a subset of original features from the dataset w/o using them.
- selection, no transformation
- methods → filter, wrapper, embedded

→ Feature extraction

- process of transforming the original features into a new set of features, either by combining or projecting them.
- methods → PCA, LDA, Autoencoders, t-SNE, UMAP

→ TF (Term frequency) → how often a word appears in a single document
if it occurs multiple times, it is impt

local impt → only within 1 document

$$TF(t, d) = \frac{\text{no. of times } t \text{ appears in } d}{\text{total terms in doc. } d}$$

→ IDF (Inverse Document Frequency) → measures how rare or common a word is across all documents.

global impt

$$IDF(t) = \log \left(\frac{N}{DF(t)} \right)$$

total no. of docs. → N
no. of docs. containing term t → $DF(t)$
to normalize the range

- multiplying (TF × IDF) helps in balancing the two opposing effects. It means that a word is impt if it appears in a document and is rare in the corpus.
- TF-IDF produces a wtd score that indicates how characteristic a term is for a document as oppose to others.